Playing a Game to Bound the Chromatic Number

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Abstract. We provide here a new method for proving several known bounds on the chromatic number of a graph in a unified manner. To do so, we view vertices of a graph as players in a strategic game. Each player has the same set of actions, which is a set of colors. In a certain profile, where each vertex has chosen a color, a vertex gets a payoff equal to zero if its color is the same as one of its neighbors. Otherwise, the vertex gets as a payoff the number of vertices that chose the same color as its own color. In a pure Nash equilibrium of such a game, no vertex can improve its payoff by deviating unilaterally. We prove that, if we start with the trivial proper coloring, where each vertex chooses its unique name as a color, then any selfish improvement sequence (i.e., a sequence of steps, where at each step a single vertex changes its color in order to improve its payoff) reaches a pure Nash equilibrium in polynomial time, and that the number of colors used by any pure Nash equilibrium satisfies four known upper bounds on the chromatic number. This implies that there is a polynomial time algorithm for computing a proper coloring of any given graph G where the number of colors used satisfies all these bounds.

1. INTRODUCTION. One of the central optimization problems in Graph Theory and Computer Science is the problem of *vertex coloring* of graphs. Given a graph $G = (V, E)$ with n vertices, assign a color to each vertex of G so that no pair of adjacent vertices gets the same color (i.e., so that the coloring produced is *proper*) and so that the total number of distinct colors used is minimized. The global optimum of vertex coloring is the *chromatic number* $\chi(G)$, defined as the minimum number of colors required to properly color the vertices of graph G.

Vertex coloring and its generalizations have a long history. It originated from the problem of coloring the countries of a map using as few colors as possible, so that no neighboring countries are colored with the same color. Vertex coloring is directly related to one of the most famous problems in Graph Theory, the *Four Color Theorem*, stating that any map can be colored using only four colors. Although it is referred to as a theorem, most mathematicians consider it as a challenging conjecture, since the Four Color Theorem has only been proven using a computer, yielding proofs that are impossible for a human to check. However, the applications of vertex coloring are much broader than this, since coloring problems arise wherever a small number of values needs to be assigned to objects under pairwise constraints. Such applications range from the problems of frequency assignment in wireless networks, job scheduling, and register allocation, to the more popular problem of solving Sudoku puzzles.

The problem of coloring a graph using the minimum number of colors is NP-hard [**9**], and the best polynomial time approximation algorithm achieves an approximation ratio of $O(n(\log\log n)^2/(\log n)^3)$ [**7**]. It is known that the chromatic number cannot be approximated to within $\Omega(n^{1-\epsilon})$ for any constant $\epsilon > 0$, unless $\mathsf{NP} \subseteq \mathsf{co\text{-}RP}$ [**6**].

Despite these negative approximation results, several upper bounds on the chromatic number have been proven in the literature; these bounds are related to various

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graph-theoretic parameters. In particular, given a graph $G = (V, E)$, let n and m denote the number of vertices and number of edges of G. Let $\Delta(G)$ denote the maximum degree of a vertex in G, and let $\Delta_2(G)$ be the maximum degree that a vertex $v \in V$ can have subject to the condition that v is adjacent to at least one vertex of degree no less than the degree of v (note that $\Delta_2(G) \leq \Delta(G)$). A *clique* in G is a subset of its vertices such that any two vertices in the subset are connected by an edge in G. The size of a clique is the number of its vertices, and the *clique number* of G, denoted by $\omega(G)$, is the maximum size of a clique in G. On the other hand, an *independent set* in G is a subset of its vertices such that no two vertices in the subset are connected by an edge in G. The size of an independent set is the number of its vertices, and the *independence number* of G, denoted by $\alpha(G)$, is the maximum size of an independent set in G. Then, it is known that

$$\chi(G) \leq \min \left\{ \Delta_2(G) + 1,\ \frac{n + \omega(G)}{2},\ n - \alpha(G) + 1,\ \frac{1 + \sqrt{1 + 8m}}{2} \right\}. \qquad (1)$$

The first bound, $\chi(G) \leq \Delta_2(G) + 1$, is due to Stacho [**13**] and implies Brooks' bound, $\chi(G) \leq \Delta(G) + 1$ [**2**]. Stacho [**13**] also gave a polynomial time algorithm that produces a coloring achieving this bound. The second bound, $\chi(G) \leq \frac{n+\omega(G)}{2}$, follows from the observation that we can color the vertices that belong to a maximum clique of G using $\omega(G)$ colors. Then, for each pair of the remaining vertices, at most one new color is needed. The third bound, $\chi(G) \leq n - \alpha(G) + 1$, follows from the fact that we can color the vertices of a maximum independent set using one color, and then color the rest $n - \alpha(G)$ vertices with $n - \alpha(G)$ new colors. The fourth bound, $\chi(G) \leq \frac{1+\sqrt{1+8m}}{2}$, follows from the fact that in an optimal coloring there must exist at least one edge between every pair of color classes, so $m \geq \frac{\chi(G)(\chi(G)-1)}{2}$.

In this article, we give a different method for proving all the above bounds on the chromatic number in a unified manner. Our method is *constructive*, in the sense that it actually computes *in polynomial time* a proper coloring using a number of colors that satisfies these bounds. In particular, we view vertices of a graph $G = (V, E)$ as players in a strategic game. Each player has the same set of actions, which is a set of $|V|$ colors. For a profile $\mathbf{c}$ (where each vertex v has chosen a color), v gets a payoff equal to zero if its color is the same as one of its neighbors. Otherwise, v gets as a payoff the number of vertices that chose the same color as v. A profile $\mathbf{c}$ is a pure Nash equilibrium of the game if no vertex can improve its payoff by deviating unilaterally.

In such a setting, if we start with the trivial proper coloring of G (where each v chooses its unique name as a color), then any selfish improvement sequence (i.e., a sequence of steps, where at each step a single vertex changes its color in order to improve its payoff) always produces a proper coloring of G. We show that such selfish improvement sequences reach an equilibrium coloring (and thus terminate, since no selfish improvements are possible in an equilibrium) in *polynomial time*; we prove this by a potential-based method [**10**]. In addition, we show that the number of colors used by any equilibrium coloring satisfies *all* bounds of the chromatic number in (1). This implies that there is a polynomial time algorithm for computing a proper coloring of any given graph G where the number of colors used satisfies all these bounds.

Graph coloring games have been studied before, but in a very different context than here. In these games there are two players, who are introduced with a graph and k colors. A legal move of either player consists of choosing an uncolored vertex v, and assigning to it any of the k colors that has not been assigned to any neighbor of v. In one variant of such a game [1] the first player which is unable to move loses the game. In another variant [1] the first player wins if and only if the game ends with all vertices colored. Further variants have also been studied by, e.g., [**8, 4**].

We note that the results presented here have originally appeared in [**12**], and have been applied to a distributed setting in [**3**]. Escoffier et al. [**5**] have extended our results by proving that the number of colors used in a Nash equilibrium does not exceed one more bound, namely

$$\frac{\chi(G)+1}{2}+\sqrt{m-(\chi(G)+1)(\chi(G)-1)/4}.$$

2. VERTEX COLORING AS A GAME.

Notation. For a finite set A we denote by $|A|$ the cardinality of A. Let $G=(V,E)$ be a simple, undirected graph with vertex set V and set of edges E. For a vertex $v \in V$ denote its set of neighbors as $N(v)=\{u \in V : \{u,v\} \in E\}$, and let $\deg(v) = |N(v)|$ be its degree. Let $\Delta(G)=\max_{v\in V}\deg(v)$ be the maximum degree of G. Let $\Delta_2(G)=\max_{u\in V}\max_{v\in N(u):d(v)\le d(u)}\deg(v)$ be the maximum degree that a vertex v can have, subject to the condition that v is adjacent to at least one vertex of degree no less than $\deg(v)$. Clearly, $\Delta_2(G)\le\Delta(G)$. Let $\chi(G)$ denote the chromatic number of G, i.e., the minimum number of colors needed to color the vertices of G such that no adjacent vertices get the same color. Let $\omega(G)$ and $\alpha(G)$ denote the clique number and independence number of G, i.e., the number of vertices in a maximum clique and a maximum independent set of G, respectively.

The graph coloring game. Given a finite, simple, undirected graph $G=(V,E)$ with $|V|=n$ vertices, we define the *graph coloring game* $\Gamma(G)$ as the game in strategic form where the set of players is the set of vertices V, and the action set of each vertex is a set of n colors $X=\{x_1,\dots,x_n\}$. A *configuration* or *pure strategy profile* $\mathbf{c}=(c_v)_{v\in V}\in X^n$ is a vector representing a combination of actions, one for each vertex. That is, c_v is the color chosen by vertex v. For a configuration $\mathbf{c}\in X^n$ and a color $x\in X$, we denote by $n_x(\mathbf{c})$ the number of vertices that are colored x in $\mathbf{c}$, i.e., $n_x(\mathbf{c})=|\{v\in V : c_v=x\}|$. The *payoff* that vertex $v\in V$ receives in the configuration $\mathbf{c}\in X^n$ is

$$\lambda_v(\mathbf{c})=\begin{cases}0 & \text{if } \exists u\in N(v): c_u=c_v\\ n_{c_v}(\mathbf{c}) & \text{else}\end{cases}.$$

A *pure Nash equilibrium* [**11**] (PNE in short) is a configuration $\mathbf{c}\in X^n$ such that no vertex can increase its payoff by deviating unilaterally. Let $(x,\mathbf{c}_{-v})$ denote the configuration resulting from $\mathbf{c}$ if vertex v chooses color x while all the remaining vertices preserve their colors. Then we make the following definition.

Definition 1. A configuration $\mathbf{c}\in X^n$ of the graph coloring game $\Gamma(G)$ is a pure Nash equilibrium, or an *equilibrium coloring*, if, for all vertices $v\in V$ and for all colors $x\in X$, $\lambda_v(x,\mathbf{c}_{-v})\le\lambda_v(\mathbf{c})$.

A vertex $v\in V$ is *unsatisfied* in the configuration $\mathbf{c}\in X^n$ if there exists a color $x\ne c_v$ such that $\lambda_v(x,\mathbf{c}_{-v})>\lambda_v(\mathbf{c})$; else we say that v is *satisfied*. For an unsatisfied vertex $v\in V$ in the configuration $\mathbf{c}$, we say that v performs a *selfish step* if v unilaterally deviates to some color $x\ne c_v$ such that $\lambda_v(x,\mathbf{c}_{-v})>\lambda_v(\mathbf{c})$.

3. EXISTENCE AND TRACTABILITY OF EQUILIBRIUM COLORINGS.

Theorem 1. *Every graph coloring game $\Gamma(G)$ possesses at least one pure Nash equilibrium, and there exists a pure Nash equilibrium $\mathbf{c}$ using $\chi(G)$ colors.*

Proof. Consider any optimal coloring $\mathbf{o} = (o_v)_{v\in V} \in X^n$ of G. Then $\mathbf{o}$ uses $\chi(G)$ colors. Without loss of generality, we assume these colors are $\{1, 2, \ldots, \chi(G)\}$. For each optimal coloring $\mathbf{o}$ consider the vector $\mathbf{L_o}$ obtained by sorting the values $n_i(\mathbf{o})$, $i = 1, \ldots, \chi(G)$, from largest to smallest. Let $\hat{\mathbf{o}}$ correspond to the lexicographically greatest vector $\mathbf{L}_{\hat{\mathbf{o}}}$. We will show that $\hat{\mathbf{o}}$ is a pure Nash equilibrium. First, since $\hat{\mathbf{o}}$ is a proper coloring, all vertices receive a payoff no less than 1, so no vertex has any incentive to choose a new color other than those already used. Now consider a vertex v which is assigned color $\hat{o}_v$ and let i be the coordinate that corresponds to $\hat{o}_v$ in $\mathbf{L}_{\hat{\mathbf{o}}}$. If v had an incentive to choose a color $x \neq \hat{o}_v$, then $n_x(\hat{\mathbf{o}}) > n_{\hat{o}_v}(\hat{\mathbf{o}})$. Thus, if v deviates, this would yield another optimal coloring $\hat{\mathbf{o}}'$ where $\mathbf{L}_{\hat{\mathbf{o}}'}$ is lexicographically greater than $\mathbf{L}_{\hat{\mathbf{o}}}$, a contradiction. Therefore $\hat{\mathbf{o}}$ is a pure Nash equilibrium using $\chi(G)$ colors. ■

Lemma 1. *Every pure Nash equilibrium $\mathbf{c}$ of $\Gamma(G)$ is a proper coloring of G.*

Proof. Assume, by contradiction, that $\mathbf{c}$ is not a proper coloring. Then there exists some vertex $v \in V$ such that $\lambda_v(\mathbf{c}) = 0$. Clearly, there exists some color $x \in X$ such that $c_u \neq x$ for all $u \in V$. Therefore $\lambda_v(x, \mathbf{c}_{-v}) = 1 > 0 = \lambda_v(\mathbf{c})$, which contradicts the fact that $\mathbf{c}$ is a pure Nash equilibrium. ■

Theorem 2. *For any graph coloring game $\Gamma(G)$, a pure Nash equilibrium can be computed in $O(n \cdot \alpha(G))$ selfish steps, where n is the number of vertices of G and $\alpha(G)$ is the independence number of G.*

Proof. We define the function $\Phi : P \to \mathbb{R}$, where $P \subseteq X^n$ is the set of all configurations that correspond to proper colorings of the vertices of G, as $\Phi(\mathbf{c}) = \frac{1}{2}\sum_{x\in X} n_x^2(\mathbf{c})$, for all proper colorings $\mathbf{c}$. Fix a *proper* coloring $\mathbf{c}$. Assume that vertex $v \in V$ can improve its payoff by deviating and selecting color $x \neq c_v$. This implies that the number of vertices colored c_v in $\mathbf{c}$ is at most the number of vertices colored x in $\mathbf{c}$, i.e., $n_{c_v}(\mathbf{c}) \leq n_x(\mathbf{c})$. If v indeed deviates to x, then the resulting configuration $\mathbf{c}' = (x, \mathbf{c}_{-v})$ is again a proper coloring (vertex v can only decrease its payoff by choosing a color that is already used by one of its neighbors, and v is the only vertex that changes its color). The improvement on v's payoff will be

$$\lambda_v(\mathbf{c}') - \lambda_v(\mathbf{c}) = n_x(\mathbf{c}') - n_{c_v}(\mathbf{c}) = n_x(\mathbf{c}) + 1 - n_{c_v}(\mathbf{c}).$$

Moreover,

$$\begin{aligned}\Phi(\mathbf{c}') - \Phi(\mathbf{c}) &= \frac{1}{2}\left(n_x^2(\mathbf{c}') + n_{c_v}^2(\mathbf{c}') - n_x^2(\mathbf{c}) - n_{c_v}^2(\mathbf{c})\right)\\ &= \frac{1}{2}\left((n_x(\mathbf{c}) + 1)^2 + (n_{c_v}(\mathbf{c}) - 1)^2 - n_x^2(\mathbf{c}) - n_{c_v}^2(\mathbf{c})\right)\\ &= n_x(\mathbf{c}) + 1 - n_{c_v}(\mathbf{c})\\ &= \lambda_v(\mathbf{c}') - \lambda_v(\mathbf{c}).\end{aligned}$$

Therefore, if any vertex v performs a selfish step (i.e., changes its color so that its payoff is increased), then the value of Φ is increased as much as the payoff of v is increased. Now, the payoff of v is increased by at least 1. So after any selfish step the value of Φ increases by at least 1. Now observe that, for all proper colorings $\mathbf{c} \in P$ and for all colors $x \in X$, $n_x(\mathbf{c}) \le \alpha(G)$. Therefore

$$\Phi(\mathbf{c}) = \frac{1}{2}\sum_{x\in X} n_x^2(\mathbf{c}) \le \frac{1}{2}\sum_{x\in X}(n_x(\mathbf{c}) \cdot \alpha(G)) = \frac{1}{2}\alpha(G)\sum_{x\in X} n_x(\mathbf{c}) = \frac{n \cdot \alpha(G)}{2}.$$

Moreover, the minimum value of Φ is $\frac{1}{2}n$. Therefore, if we allow any unsatisfied vertex (but only one each time) to perform a selfish step, then after at most $\frac{n\cdot\alpha(G)-n}{2}$ steps there will be no vertex that can improve its payoff. This is because Φ will have reached a local maximum, which is no more than the global maximum, which is no more than $(n \cdot \alpha(G))/2$. This implies that a pure Nash equilibrium will have been reached. Of course, we have to start from an initial configuration that is a proper coloring so as to ensure that the procedure will terminate in $O(n \cdot \alpha(G))$ selfish steps; this can be found easily since there is always the trivial proper coloring that assigns a different color to each vertex of G. ■

4. BOUNDS ON THE NUMBER OF COLORS.

Proposition 1. *In any pure Nash equilibrium* $\mathbf{c}$ *of* $\Gamma(G)$*, if* $n_{c_v}(\mathbf{c}) \le n_x(\mathbf{c})$ *for some vertex* v *and some color* $x \ne c_v$*, then* v *is adjacent to some vertex colored* x.

Proof. Assume by contradiction that v has no neighbor colored x. Then, since $\mathbf{c}$ is a pure Nash equilibrium, it must hold that $n_{c_v}(\mathbf{c}) \ge n_x(\mathbf{c}) + 1$, implying that $n_{c_v}(\mathbf{c}) \ge n_{c_v}(\mathbf{c}) + 1$, a contradiction. ■

Lemma 2. *In any pure Nash equilibrium* $\mathbf{c}$ *of* $\Gamma(G)$*, the number* k *of total colors used satisfies* $k \le \Delta_2(G) + 1$ *and hence* $k \le \Delta(G) + 1$.

Proof. Let k be the total number of distinct colors used in $\mathbf{c}$. If $k = 1$ then it is easy to observe that G must be totally disconnected, i.e., $\Delta(G) = \Delta_2(G) = 0$ and therefore $k = \Delta_2(G) + 1$. Now assume $k \ge 2$. Let $x_i, x_j \in X$ be the two colors used in $\mathbf{c}$ that are assigned to the minimum number of vertices. Without loss of generality, assume that $n_{x_i}(\mathbf{c}) \le n_{x_j}(\mathbf{c}) \le n_x(\mathbf{c})$ for all colors $x \notin \{x_i, x_j\}$ used in $\mathbf{c}$. Let v be a vertex such that $c_v = x_i$. The payoff of vertex v is $\lambda_v(\mathbf{c}) = n_{x_i}(\mathbf{c})$. Now consider any other color $x \ne x_i$ that is used in $\mathbf{c}$. By Proposition 1, there is an edge between vertex v and at least one vertex of color x. Hence the degree of vertex v is at least the total number of colors used minus 1, i.e., $\deg(v) \ge k - 1$. Furthermore, let u be a vertex of color $c_u = x_j$ that v is connected to. Similarly, u must be connected to at least one vertex of color x, for all $x \notin \{x_i, x_j\}$ used in $\mathbf{c}$. Moreover, u is also connected to v. Therefore $\deg(u) \ge k - 1$. Now,

$$\begin{aligned}
\Delta_2(G) &= \max_{s\in V} \max_{\substack{t\in N(s)\\ \deg(t)\le\deg(s)}} \deg(t) \\
&\ge \max\left\{ \max_{\substack{t\in N(v)\\ \deg(t)\le\deg(v)}} \deg(t), \max_{\substack{t\in N(u)\\ \deg(t)\le\deg(u)}} \deg(t) \right\} \\
&\ge \min\{\deg(u), \deg(v)\} \\
&\ge k-1
\end{aligned}$$

and therefore $k \le \Delta_2(G) + 1$ as needed. ■

Lemma 3. *In a pure Nash equilibrium* $\mathbf{c}$ *of* $\Gamma(G)$*, all vertices that are assigned unique colors form a clique.*

Proof. Assume that the colors c_v and c_u chosen by vertices v and u are unique, i.e., $n_{c_v}(\mathbf{c}) = n_{c_u}(\mathbf{c}) = 1$. Then the payoff for both vertices is 1. If there is no edge between u and v then, since $\mathbf{c}$ is an equilibrium, it must hold that $1 = \lambda_v(\mathbf{c}) \geq \lambda_v(c_u, \mathbf{c}_{-v}) = 2$, a contradiction. ■

Lemma 4. *In any pure Nash equilibrium* $\mathbf{c}$ *of* $\Gamma(G)$*, the number k of total colors used satisfies* $k \leq \frac{n+\omega(G)}{2}$.

Proof. Assume there are $t \geq 0$ vertices that are each assigned a unique color. These t vertices form a clique (Lemma 3), hence $t \leq \omega(G)$. The remaining $n - t$ vertices are assigned non-unique colors, so the number of colors in $\mathbf{c}$ is $k \leq t + \frac{n-t}{2} = \frac{n+t}{2} \leq \frac{n+\omega(G)}{2}$. ■

Lemma 5. *In any pure Nash equilibrium* $\mathbf{c}$ *of* $\Gamma(G)$*, the number k of total colors used satisfies* $k \leq \frac{1+\sqrt{1+8m}}{2}$.

Proof. Without loss of generality, assume that the k colors used in $\mathbf{c}$ are $x_1, \ldots, x_k$. Let V_i, $1 \leq i \leq k$, denote the subset of all vertices $v \in V$ such that $c_v = x_i$. Without loss of generality, assume that $|V_1| \leq |V_2| \leq \cdots \leq |V_k|$. By Proposition 1, for each vertex $v_i \in V_i$, there is an edge between v_i and some $v_j \in V_j$, for all $j > i$. This implies that $m \geq \sum_{i=1}^{k-1} |V_i|(k-i)$ and, since $|V_i| \geq 1$ for all $i \in \{1, \ldots, k\}$, $m \geq \sum_{i=1}^{k-1}(k-i)$ or equivalently $m \geq \frac{k(k-1)}{2}$ or equivalently $k^2 - k - 2m \leq 0$, which implies $k \leq \frac{1+\sqrt{1+8m}}{2}$. ■

Theorem 3. *In any pure Nash equilibrium* $\mathbf{c}$ *of* $\Gamma(G)$*, the number k of total colors used satisfies* $k \leq n - \alpha(G) + 1$.

Proof. Let t be the maximum payoff of a vertex in $\mathbf{c}$, i.e., $t = \max_{x \in X} n_x(\mathbf{c})$. Partition the set of vertices into t sets $V_1, \ldots, V_t$ so that $v \in V_i$ if and only if $\lambda_v(\mathbf{c}) = i$ (note that each vertex appears in exactly one such set, however not all sets have to be nonempty). Let k_i denote the total number of colors that appear in V_i. Clearly, $|V_i| = i \cdot k_i$ and the total number of colors used in $\mathbf{c}$ is $k = \sum_{i=1}^{t} k_i$. Now consider a maximum independent set I of G. The vertices in V_1 have payoff equal to 1, therefore they are assigned unique colors, so, by Lemma 3, the vertices in V_1 form a clique. Therefore I can only contain at most one vertex among the vertices in V_1. Our goal is to give an upper bound for the size of I. First we prove the following:

Claim 1. If there exists some $i > 1$ such that $k_i = 1$ and I contains all the vertices in V_i, then $k \leq n - \alpha(G) + 1$.

Proof of Claim 1. Let x denote the unique color that appears in V_i. Since I contains all the vertices in V_i, then it cannot contain any vertex in $V_1 \cup \cdots \cup V_{i-1}$. This is so because each vertex $v \in V_j$, $j < i$, is connected by an edge with at least one vertex of color x (Proposition 1). Furthermore, each vertex in V_i has at least one neighbor of each color that appears in $V_{i+1} \cup \cdots \cup V_t$. Therefore

$$\alpha(G) = |I| \leq |V_i| + \sum_{j=i+1}^{t} |V_j| - \sum_{j=i+1}^{t} k_j = n - \sum_{j=1}^{i-1} |V_j| - k + \sum_{j=1}^{i} k_j$$

which gives

$$k \le n - \alpha(G) + \sum_{j=1}^{i-1} (k_j - |V_j|) + k_i \le n - \alpha(G) + k_i = n - \alpha(G) + 1. \qquad \blacksquare$$

So now it suffices to consider the case where, for all $i > 1$, if $k_i = 1$, I does not contain all the vertices in V_i, i.e., I contains at most $|V_i| - 1 = |V_i| - k_i$ vertices that belong to V_i. In order to complete the proof we need the following.

Claim 2. For all $i > 1$ with $k_i \neq 1$, I cannot contain more than $|V_i| - k_i$ vertices among the vertices in V_i.

Proof of Claim 2. This is clearly true for $k_i = 0$ (and hence $|V_i| = 0$). Now assume that $k_i \ge 2$. Observe that, for all vertices $v_i \in V_i$ there must exist an edge between v_i and a vertex of each one of the remaining $k_i - 1$ colors that appear in V_i (Proposition 1). Fix a color x of the k_i colors that appear in V_i. If I contains all vertices of color x, then it cannot contain any vertex of any color other than x that appears in V_i. Therefore I can contain at most $i \le (i-1)k_i = |V_i| - k_i$ vertices among the vertices in V_i. On the other hand, if I contains at most $i - 1$ vertices of each color x that appears in V_i, then I contains again at most $(i-1)k_i = |V_i| - k_i$ vertices among the vertices in V_i. $\blacksquare$

Therefore I cannot contain more than $|V_i| - k_i$ vertices among the vertices of V_i, for all $i > 1$, plus one vertex from V_1. Therefore:

$$|I| = \alpha(G) \le 1 + \sum_{i=2}^{t} (|V_i| - k_i) = 1 + n - |V_1| - (k - |V_1|) = n - k + 1.$$

So, in any case, $k \le n - \alpha(G) + 1$ as needed. $\blacksquare$

The bounds given by Lemmata 2, 4, 5 and Theorem 3, together with the facts that any Nash equilibrium is a proper coloring (Lemma 1) and that a Nash equilibrium can be computed in polynomial time (Theorem 2), imply the following.

Corollary 1. *For any graph G, a proper coloring that uses at most*

$$k \le \min \left\{ \Delta_2(G) + 1,\ \frac{n + \omega(G)}{2},\ \frac{1 + \sqrt{1 + 8m}}{2},\ n - \alpha(G) + 1 \right\}$$

colors can be computed in polynomial time.

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